Roan Alexander James

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RESEARCH INTERESTS

Probability Theory, Stochastic Processes, Directed Polymer Models, Stochastic Partial Differential Equations, and the Kardar–Parisi–Zhang (KPZ) Universality Class.

EDUCATION

University of Rochester Rochester, NY
PhD in Mathematics *Sep 2022 – Expected May 2027*

- Advisor: Professor Arjun Krishnan.
- Research focused on directed polymer models, stochastic heat equations, and the KPZ universality class.

University of Sheffield Sheffield, UK
Master of Mathematics (MMath), First Class Honours *Sep 2017 – Jul 2021*

- Awarded the T. M. Flett Prize for the highest mark in Pure Mathematics.

PREPRINTS

- R. A. James and A. Krishnan. *Directed Polymers in an External Field at Intermediate Disorder.* In preparation; expected on arXiv in November 2026.

AWARDS AND HONORS

- **T. M. Flett Prize**, University of Sheffield 2021
  Awarded for the highest mark in Pure Mathematics.

TEACHING EXPERIENCE

University of Rochester Rochester, NY

Instructor of Record

Term	Course	Course Title	Format
Fall 2026*	MATH 162	Calculus II	In Person
Summer 2026	MATH 165	Linear Algebra and Differential Equations	Online
Summer 2025	MATH 201	Introduction to Probability	Online
Summer 2024	ECO	Early Connection Opportunity: Basics of Mathematics	In Person
Summer 2023	ECO	Early Connection Opportunity: Basics of Mathematics	In Person

*Current semester.

Workshop Instructor and Teaching Assistant

Term	Course	Course Title	Role
Spring 2026	MATH 143	Calculus III	Workshop Instructor
Fall 2025	MATH 265	Abstract Algebra	Grader
Spring 2025	MATH 217	Politics and Mathematics	Grader
Fall 2024	MATH 503	Graduate Probability	Grader
Spring 2024	MATH 310	Undergraduate Game Theory	Grader
Fall 2023	MATH 162	Calculus II	Workshop Instructor
Fall 2023	MATH 202	Undergraduate Complex Analysis	Grader
Spring 2023	MATH 143	Calculus III	Workshop Instructor
Fall 2022	MATH 164	Multidimensional Calculus	Workshop Instructor

Student Evaluations

Across four semesters of workshops, students rated my teaching **4.57/5 overall** (58 respondents). Scores are on a five-point scale; the overall average is weighted by the number of respondents (N) in each section.

Section	N	Teaching	Speaking	In-Class Access	Outside Access	Overall
MATH 164 — Multidimensional Calculus, Fall 2022						
Thu. 6:15–7:30 PM	10	4.30	4.40	4.60	4.50	**4.50**
Fri. 3:25–4:40 PM	9	4.78	4.78	4.67	4.22	**4.78**
Fri. 2:00–3:15 PM	9	4.78	4.67	4.89	5.00	**4.78**
MATH 143 — Calculus III, Spring 2023						
Thu. 3:25–4:40 PM	2	4.50	4.50	4.50	5.00	**4.50**
Wed. 12:30–1:45 PM	3	4.67	4.67	5.00	4.00	**4.67**
Wed. 2:00–3:15 PM	4	4.75	4.75	4.75	4.25	**4.75**
MATH 162 — Calculus II, Fall 2023						
Fri. 3:25–4:40 PM	9	3.78	4.56	4.33	3.62	**4.44**
Fri. 4:50–6:05 PM	2	4.50	5.00	5.00	4.00	**4.50**
MATH 143 — Calculus III, Spring 2026						
Thu. 3:25–4:40 PM	5	4.60	4.80	4.60	4.20	**4.60**
Tue. 2:00–3:15 PM	4	3.75	4.50	4.50	3.75	**4.00**
Fri. 10:25–11:40 AM	1	4.00	5.00	2.00	5.00	**4.00**
All workshops	**58**					**4.57**

"He takes the time to show the reasoning behind everything and to help us understand why we are doing what we're doing." — MATH 143, Spring 2023

"He would ask us what we don't understand or don't have confidence in, then prepared questions corresponding to it." — MATH 164, Fall 2022

Further student comments are available at roanalexanderjames.github.io.

TALKS AND LECTURES

Summer Lecture Series: Introduction to Polymer Models

A six-lecture introductory series on directed polymers in a random environment for graduate students with a background in basic probability.

- **Lecture 1: Directed Polymers in a Random Environment**
  Summer Lecture Series, University of Rochester

The polymer measure on nearest-neighbour paths, the random environment $\{\omega(n,x)\}$, the partition function, and motivation from statistical mechanics.

- **Lecture 2: The Partition Function as a Martingale**
  Summer Lecture Series, University of Rochester
  The normalised partition function, its martingale property with respect to the environment filtration, and the almost-sure limit.

- **Lecture 3: The Zero–One Law and the Phase Transition**
  Summer Lecture Series, University of Rochester
  The weak- and strong-disorder dichotomy and the critical inverse temperature β_c.

- **Lecture 4: Weak Disorder and L^2 Techniques**
  Summer Lecture Series, University of Rochester
  Second moment computations, replica overlap, and uniform integrability in dimensions $d \geq 3$.

- **Lecture 5: Strong Disorder, Localisation, and Low Dimensions**
  Summer Lecture Series, University of Rochester
  Strong disorder in dimensions $d = 1, 2$, localisation of the path measure, and the role of free energy.

- **Lecture 6: Scaling Limits and Connections to the KPZ Class**
  Summer Lecture Series, University of Rochester
  Intermediate disorder scaling, convergence to the stochastic heat equation, and an outlook on KPZ universality.

Graduate Student Seminar: The Stochastic Heat Equation

- **The Stochastic Heat Equation, Part I: Construction and Mild Solutions**
  Graduate Student Seminar, University of Rochester
  Space–time white noise, the multiplicative stochastic heat equation, Walsh stochastic integration, and existence and uniqueness of mild solutions via Picard iteration.

- **The Stochastic Heat Equation, Part II: Chaos Expansion and the KPZ Connection**
  Graduate Student Seminar, University of Rochester
  The Wiener chaos expansion, positivity and moment bounds, the Hopf–Cole transform linking the stochastic heat equation to KPZ, and intermittency.

Probability Seminar and Workshops

- **Simple Random Walks and Discrete Stochastic Differential Equations**
  Vertical Integration Workshop, University of Rochester April 18, 2026
  Simple random walks in random environments and their connection to stochastic partial differential equations, with scaled partition functions interpreted as discrete analogues of solutions to the Parabolic Anderson Model. Joint work with Arjun Krishnan.

- **The Second Moment Method for the Partition Function**
  Probability Seminar, University of Rochester
  The Paley–Zygmund inequality, second moment calculations in terms of random-walk overlap, weak disorder in transient dimensions, and limitations of the method in low dimensions.

CONFERENCE AND SCHOOL ATTENDANCE

Date	Conference or School	Location	Financial Support
June 2026	MSRI Summer School: Random Growth Models and Hamilton–Jacobi Equations	University of California, Berkeley	Accommodation and $850 in travel expenses
November 2025	Northeast Probability Seminar	New York University, New York	$700 in travel expenses
July 2025	Seminar on Stochastic Processes 2025	Union College, Schenectady, New York	Accommodation and $250 in travel expenses
June 2025	School on Interacting Particle Systems and Random Matrix Theory	Alfréd Rényi Institute, Budapest	Accommodation and $1,000 in travel expenses
November 2024	Northeast Probability Seminar	Columbia University, New York	$700 in travel expenses
October 2024	44th Midwest Probability Conference	Northwestern University, Evanston, Illinois	$750 in travel expenses
September 2024	Conference on New Developments in Probability	Centre de Recherches Mathématiques, Montreal	$1,000 in travel expenses

SERVICE, LEADERSHIP, AND MENTORING

Departmental and Professional Leadership

- **President, AMS Graduate Student Chapter**
  American Mathematical Society Student Chapter, University of Rochester
  Organise academic and professional-development events, coordinate with the national AMS and the department on funding and logistics, communicate conferences, fellowships, and travel opportunities, and support community-building across the graduate cohort. Events organised include:
  - *Careers Panel for Mathematics Graduate Students* September 18, 2026
  - *Panel Discussion: The Future of Mathematics with AI* October 9, 2026

- **Organiser, Graduate Student Seminar**
  Department of Mathematics, University of Rochester
  Manage the seminar schedule, recruit speakers, coordinate room bookings and announcements, and help graduate speakers prepare talks for a broad mathematical audience.

- **Weekly Mathematics Study Hall**
  Department of Mathematics, University of Rochester
  Answer questions from students in any undergraduate mathematics course on their coursework and homework.

Mentoring and Undergraduate Research

- **Undergraduate Reading Course: Combinatorics of the Simple Random Walk** Fall 2026
  Department of Mathematics, University of Rochester
  Lead a reading course for five undergraduate students on the combinatorics of the simple random walk.

- **Research Mentor, Undergraduate Research Project**
  Liam Hillis, University of Rochester
  Supervise an undergraduate project on directed polymer models in random environments, in-

cluding a reading programme on polymer measures, partition functions, martingale convergence, second moment methods, and random-walk overlap estimates.

PROFESSIONAL EXPERIENCE

Portland Training Sheffield, UK
Administrative Assistant *Aug 2021 – Aug 2022*

- Managed complex datasets using Excel and coordinated customer interactions.
- Achieved two promotions within eight months as responsibilities increased.

TECHNICAL SKILLS

- **Languages & Tools:** LaTeX, Python, R / RStudio, Microsoft Excel.
- **Mathematical Areas:** Probability Theory, Stochastic Processes, Directed Polymers, Stochastic PDEs, and financial applications of probability.